

1. Define how java works. Write a program to demonstrate it.

Institute of Engineering & Technology
BE II year Information Technology 3ITRC3 Data Structures
Class Test 1 August 2023

MM - 20

Duration : 70 Mins

Note: All questions are compulsory. Make suitable assumptions, if necessary.

- ① | ② | ③
1. Write a procedure/algorithm/code for multiplication of two polynomials. What is the computing time of your procedure? 3+1
2. An $m \times n$ matrix has a saddle point if some entry $A(i,j)$ is the smallest value in row i and the largest value in column j . Write a procedure/algorithm/code which determines the location of a saddle point if one exists. What is the computation time? 3+1
3. You are given two sorted arrays. You need to store the intersection of the arrays in a new sorted array. Write the procedure/algorithm/code for the same. 3
4. Differentiate between Abstract Data Type and Data Structures. Write an ADT for Ordered List. 2+3
5. What is Binary Search? Calculate the time complexity of it using apriori estimate. 1+3